Supplementary Material for RelationalRAG-Tables: A Learned Join Verifier for QA over Tables and Knowledge Graphs

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I. Scope of this Supplement

This document collects the *supporting* stress tests behind the main paper's regime characterization. The main paper is self-contained, and nothing here is required to follow its argument. We gather these tables for two reasons: to give a table anchor to numbers the main paper reports inline, and to expose per-seed and per-regime detail that does not fit the page budget. Every number is a measured result from the released harness, and we report seeds explicitly and add no value we did not run. Section and table references of the form "main paper, Table N" point to the primary submission.

II. Main-Text Tables Replaced by Plots

The main paper promotes five dense result tables into compact figures. Tables I–IV and Table V preserve the exact numeric values behind those plots.

TABLE I
End-to-end exact match by verifier and task family on the drifted synthetic corpus (1,392 questions, 60% join drift). "Abst." is the number of abstentions. Oracle is an upper bound with gold join labels.

Verifier	Single	Join	Multi	Back	Abst.	EM	Abst.
Exact-match	1.00	0.43	0.43	0.43	**1.00**	0.569	636
No verifier	1.00	**1.00**	**1.00**	0.40	0.00	0.860	0
Learned	1.00	0.93	0.93	**0.58**	0.08	0.856	50
Deep ensemble	1.00	0.94	0.94	0.57	0.08	**0.861**	47
Oracle (UB)	1.00	1.00	1.00	1.00	1.00	1.000	36

TABLE II
Heterogeneous-drift join (one ICD-10 KG, each record drifting by hierarchy, OCR-corruption, *or* cross-vocabulary), recall@1, mean ± sd over five seeds.

Method	recall@1 (mean ± sd)
Exact code	0.011 ± 0.009
Fuzzy (edit)	0.251 ± 0.023
Prefix / hierarchy rule	0.258 ± 0.030
Embedding	0.689 ± 0.035
Learned verifier	$\mathbf{0.752 \pm 0.035}$

III. ICD-10 + Wikidata: OCR-Corruption Sweep

The main paper states that under data-entry corruption the prefix-hierarchy rule collapses while embedding on the surviving description overtakes it. Table VI makes the collapse explicit as a sweep over the fraction of drifted records whose codes are OCR-corrupted, at a single seed for internal consistency. The prefix rule's drift recall@1 falls $0.729 \rightarrow 0.397$ as corruption goes $0 \rightarrow 1$, while embedding, which ignores the code, is flat and becomes the strongest single signal once codes are fully corrupted. The learned verifier tracks whichever signal is stronger but *never significantly beats the prefix rule* (paired bootstrap $p > 0.45$ at every level), which is the cautionary finding of the main paper shown here across the full corruption range rather than at one point.

TABLE III
BIRD value linking: recall@1 on held-out instances (387 test, 38 drifted).

Scorer	recall@1 (all)	recall@1 (drift)
Exact match	0.788	0.000
Fuzzy match	0.674	0.237
Embedding cosine	0.881	0.868
Learned verifier	**0.928**	**0.921**

TABLE IV
End-to-end Execution Accuracy on BIRD dev (local 7B-Coder).

Pipeline	EX	EX (drift)
Ablation (n=800, 72 drift)		
baseline (full schema)	0.458	0.292
+ column values	0.476	0.333
+ value hints (all)	0.446	0.347
+ value hints (selective)	0.456	0.319
+ schema pruning	0.479	0.319
+ pruning + values	**0.483**	**0.361**
Full dev (n=1534, 131 drift)		
baseline	0.454	0.328
+ pruning + values	**0.477**	**0.420**

IV. Heterogeneous Drift: Per-Seed and Per-Type Detail

The main paper reports that on a column mixing hierarchy, OCR, and cross-vocabulary drift, a single learned verifier beats every fixed baseline. Table VII gives the five-seed breakdown behind that mean. The verifier beats the prefix rule decisively in all five seeds ($\Delta \approx +0.49$, $p < 10^{-3}$ throughout) and beats embedding in three of five (seeds 1, 2, and 3), while on seeds

TABLE V
CROSS-VOCABULARY JOIN (ICD-9 TABLE → ICD-10 KG, 129 HELD-OUT), RECALL@1.

Method	recall@1
Exact code	0.008
Fuzzy code	0.008
Prefix rule	0.008
Embedding	**0.907**
Learned verifier	0.899

TABLE VI
ICD-10 JOIN, *drift-subset* RECALL@1 AS A FUNCTION OF CODE-CORRUPTION FRACTION (SEED 0, WITH 59 TO 69 DRIFTED TEST RECORDS). p IS A PAIRED BOOTSTRAP OF VERIFIER VS. PREFIX.

Corrupt frac.	Exact	Fuzzy	Prefix	Embed	Verifier
0.0 (clean)	0.000	0.644	**0.729**	0.458	0.746
0.5	0.015	0.493	**0.594**	0.493	0.580
1.0 (full)	0.032	0.365	0.397	**0.460**	0.413
Verifier vs. prefix: n.s. throughout (p=0.59, 0.86, 0.45)					

0 and 4 the embedding gap is positive but not significant. The win over the strongest single baseline is therefore real but modest, exactly as the main paper claims.

TABLE VII
HETEROGENEOUS-DRIFT JOIN, PER-SEED RECALL@1 (ONE ICD-10 KG, WITH EACH RECORD DRIFTING BY HIERARCHY, OCR, OR CROSS-VOCABULARY). Δ AND p ARE VERIFIER VS. EMBEDDING (PAIRED BOOTSTRAP, 10^4 RESAMPLES).

Seed	Exact	Fuzzy	Prefix	Embed	Verifier	Δ_{emb}	p
0	0.022	0.256	0.261	0.678	0.700	+0.022	0.257
1	0.006	0.228	0.256	0.733	0.794	+0.061	0.016
2	0.022	0.283	0.311	0.672	0.772	+0.100	<0.001
3	0.000	0.222	0.222	0.639	0.772	+0.133	<0.001
4	0.006	0.267	0.239	0.722	0.722	+0.000	0.532
mean	0.011	0.251	0.258	0.689	**0.752**	+0.063	3/5 sig.

The per-drift-type breakdown that explains *why* the verifier wins the mixture, namely that it is never the single best method on any one drift type yet inherits the better signal per record, now appears in the main text as part of the heterogeneous-drift result.

V. ROBUSTNESS: CROSS-REGIME TRANSFER AND DESCRIPTION NOISE

Two robustness probes support the claim that the verifier leans on whichever feature survives. First, *transfer*: trained only on clean hierarchy drift, the verifier recovers OCR-corrupted codes zero-shot at recall@1 0.528 on 36 held-out corrupted records, as good as the 0.472 from training directly on corruption, because the learned edit and embedding features generalize across drift types. Second, *description noise*: under 50% word-dropout of descriptions (Table VIII), the verifier's drift recall@1 falls only 0.746→0.712 while embedding falls 0.458 → 0.424, and the prefix rule is unaffected because it never reads the description. The verifier degrades gracefully by shifting weight onto the noise-free code features.

TABLE VIII
DESCRIPTION-NOISE SENSITIVITY: DRIFT-SUBSET RECALL@1 UNDER 50% WORD-DROPOUT OF RECORD AND NODE DESCRIPTIONS (SEED 0).

Method	clean desc.	noisy (50% drop)
Prefix / hierarchy rule	0.729	0.729
Embedding	0.458	0.424
Learned verifier	**0.746**	**0.712**

VI. A STRONGER MATCHER: FINE-TUNED CROSS-ENCODER

A natural worry is that the feature-MLP verifier is too weak to settle the "learned vs. rule" question. We replace it with a fine-tuned MiniLM cross-encoder that reads code and description jointly (Table IX). On BIRD value linking it is excellent (0.95 overall and 0.95 on drift), beating the feature MLP, but on HybridQA it loses to lexical matching (0.205 < 0.235), and on the ICD hierarchy task (main paper) it still ties the prefix rule (p≈0.89). A stronger model does not change the regime characterization: capacity is not the operative variable.

TABLE IX
FINE-TUNED CROSS-ENCODER (MINILM) RECALL@1.

Task (held-out)	Baseline	Cross-encoder
BIRD value linking, all (387)	n/a	**0.954**
BIRD value linking, drift (38)	n/a	**0.947**
HybridQA top-1 (395)	0.235 (lexical)	0.205

VII. HYBRIDQA: FULL RETRIEVAL AND RERANKING

HybridQA is the main paper's negative result: with gold cell-to-passage links and no identifier drift, locating the answer-bearing unit is lexically dominated. Table X gives the retrieval curve. Lexical overlap dominates dense `bge-small` retrieval at every cutoff (recall@20 0.826 vs. 0.647), and a learned evidence verifier, although it improves binary discrimination (area under the ROC curve, AUROC, 0.610 vs. 0.505 and 0.493 for the dense and lexical signals alone), *loses* at top-1 reranking (0.091 vs. lexical's 0.249, and neither cascade in the main paper's cascade table recovers it). String matching is the right tool here, and we report it.

TABLE X
HYBRIDQA RETRIEVAL RECALL@k (∼397 DEV QUESTIONS). TOP-1 RERANKING AND VERIFIER AUROC ARE REPORTED IN THE TEXT.

Retrieval	@1	@5	@10	@20
Dense (`bge-small`)	0.129	0.390	0.511	0.647
Lexical	**0.244**	**0.544**	**0.670**	**0.826**
Hybrid	0.234	0.474	0.647	0.778

VIII. BIRD Value Linking: Discrimination and Selective Prediction

Beyond the recall@1 table in the main paper, the verifier's pair-classification is near-perfect and well calibrated. On the held-out split (seed 4, 387 test, 45 drifted) the verifier reaches recall@1 0.943 overall and 0.911 on drift, consistent with the main paper's 0.93 and 0.92 and the five-seed 0.892 ± 0.045 on drift. Its discrimination AUROC is 0.994, far above embedding cosine (0.824), and selective prediction by ensemble uncertainty raises accuracy on the answered fraction from 0.994 at full coverage to 1.000 at 50% (Table XI). On real value linking the verifier is both accurate and confident-when-right.

TABLE XI
BIRD value linking, held-out (seed 4). Pair-classification AUROC by scorer, and ensemble selective-prediction accuracy vs. coverage.

AUROC	verifier	exact	fuzzy	embed-cos	
	0.994	0.931	0.904	0.824	
Coverage	1.00	0.90	0.80	0.70	0.50
Accuracy	0.994	0.999	1.000	1.000	**1.000**

IX. End-to-End Execution Accuracy: Scale and Full Dev

The main paper's end-to-end gain holds at full-dev scale and at larger model size (Table XII). At full dev (n=1534) the schema-pruning-plus-values pipeline lifts EX $0.454 \rightarrow 0.477$ (+2.3), with the drift subset gaining +9.2 ($0.328 \rightarrow 0.420$). A 14B coder improves identically in relative terms (+2.3 EX), confirming the gain is not an artifact of the 7B model. Absolute EX stays below cloud-scale systems, so the contribution is the relative effect and the schema-linking diagnosis.

TABLE XII
End-to-end Execution Accuracy (BIRD dev, local coder).

Setting (n)	EX base	EX +p+v	Δ	Δ drift
7B, full dev (1534)	0.454	**0.477**	+2.3	+9.2
14B (400)	0.490	**0.513**	+2.3	+6.7

X. Calibration, System-Level Selective Prediction, and Retrieval Noise

These diagnostics answer questions the controlled study did not originally report: calibration error, selective prediction at the level of QA answers rather than pairwise joins, and the value of gating once retrieval is no longer idealized.

a) End-to-end selective prediction.: Beyond pairwise joins, we rank each answered QA question by the uncertainty of its committed join and defer the most-uncertain tail (Table XIII). System-level exact match rises from 0.889 at full coverage to 1.000 at 50% coverage. The curve is noisier than the pairwise one, because entropy is a weak error predictor on the aleatoric joins and because deterministic single-table lookups occupy the high-confidence region, but the endpoints give a clean system-level accuracy-coverage operating curve.

TABLE XIII
End-to-end (system-level) selective prediction: exact match on the answered fraction of QA questions after deferring the most-uncertain committed joins (1,345 answered).

Coverage	1.00	0.90	0.80	0.70	0.50
System EM	0.889	0.890	0.876	0.913	**1.000**

b) Retrieval noise.: The synthetic study places the gold node in the candidate pool, which inflates the no-verifier baseline. We stress this on bad-join-backtrack questions, where retrieval surfaces a plausible wrong neighbor, by dropping the gold candidate with probability p (Table XIV). The no-verifier planner commits the wrong neighbor and returns a silently wrong answer for 0.60 of questions at every p, and it never abstains. The verifier holds the wrong-answer rate lower and converts the growing pool of unrecoverable joins into explicit abstentions, whose rate climbs from 0.02 at p=0 to 0.29 at p=1. The value of gating grows precisely as retrieval degrades, which aggregate exact match on idealized retrieval hides. The p=0 row reconciles exactly with the backtrack column of the main paper (no-verifier EM 0.40, ensemble 0.57).

TABLE XIV
Retrieval-noise robustness on bad-join-backtrack questions (264 questions), dropping the gold candidate with probability p. NV = no-verifier, Ens = ensemble.

	No-verifier			Ensemble		
p	EM	wrong	abst	EM	wrong	abst
0.00	0.40	0.60	0.00	**0.57**	0.41	0.02
0.25	0.40	0.60	0.00	0.51	0.42	0.06
0.50	0.40	0.60	0.00	0.43	0.44	0.13
0.75	0.40	0.60	0.00	0.34	0.45	0.20
1.00	0.40	0.60	0.00	0.25	0.46	**0.29**

c) Faithfulness.: On the same QA run, answers are *faithful* (grounded in retrieved evidence or a deterministic traversal) for 0.94 of non-abstained questions. The deterministic planner cannot fabricate, and the residual gap is retrieval coverage on single-table lookups rather than hallucination.

d) Cost-aware utility.: Accuracy-coverage curves understate the deployment value of abstention, so we score a correct answer +1, a wrong answer $-\lambda$, and an abstention 0, and find the utility-maximizing coverage (Table XV). When wrong answers are cheap (λ=1) the optimum answers everything. When a wrong answer costs five times a correct one (λ=5), the high-stakes regime the verifier targets, the optimum is to abstain on the most-uncertain 40% of answers, which raises utility from 0.33 at full coverage to 0.60 at coverage 0.6. Abstention pays off precisely when errors are costly.

TABLE XV
COST-AWARE UTILITY ON THE SYNTHETIC QA SET. A CORRECT ANSWER SCORES $+1$, A WRONG ANSWER $-\lambda$, AND AN ABSTENTION 0. ABSTENTION BECOMES UTILITY-OPTIMAL AS WRONG ANSWERS GROW COSTLY.

Wrong-answer penalty λ	Best coverage	Utility (best / full)
1	1.0	0.78 / 0.78
2	1.0	0.67 / 0.67
5	0.6	0.60 / 0.33

XI. THRESHOLD SENSITIVITY, COMPUTE-MATCHED UNCERTAINTY, KG INCOMPLETENESS, AND EFFICIENCY

These diagnostics answer further reviewer requests: threshold stability, whether the deep ensemble earns its compute, robustness to graph incompleteness, a deployment-oriented latency breakdown, and the cost of gating.

a) Threshold sensitivity.: Table XVI sweeps the acceptance threshold τ_v. Precision rises and recall falls monotonically, and F1 is flat (0.75 to 0.77) across $\tau_v \in [0.4, 0.6]$, so the fixed τ_v=0.5 sits in a stable band near the F1 optimum rather than on a cliff.

TABLE XVI
VERIFIER DISCRIMINATION ON THE HELD-OUT SPLIT AS THE ACCEPTANCE THRESHOLD τ_v VARIES.

τ_v	0.3	0.4	0.5	0.6	0.7
Precision	0.657	0.696	0.767	0.808	0.883
Recall	0.923	0.869	0.738	0.702	0.500
F1	0.768	**0.773**	0.752	0.751	0.639

b) Does the ensemble earn its compute?: We compare the five-member ensemble against two single-model signals at one-fifth the compute: the entropy of one member and a score-margin heuristic, which rank identically (Table XVII). For *selective prediction* the single model matches the ensemble (both reach 0.86 at 50% coverage, and the single model is in fact slightly higher at full coverage, mirroring its higher raw accuracy in the main paper). The ensemble's value is therefore *calibration* (ECE 0.027 vs. the single model's 0.036, in the main-text calibration table), not selective-prediction ranking, which is consistent with the saturating K-sweep in the main paper. A practitioner who needs only ranking can use a single model.

TABLE XVII
COMPUTE-MATCHED SELECTIVE PREDICTION: ACCURACY ON THE ANSWERED FRACTION, ENSEMBLE ($5\times$) VS. A SINGLE-MODEL SIGNAL ($1\times$).

Coverage	1.00	0.90	0.80	0.70	0.50
Ensemble ($5\times$)	0.725	0.757	0.783	0.803	0.856
Single model ($1\times$)	0.743	0.755	0.779	0.793	0.861

c) KG incompleteness.: Removing a fraction of graph edges (Table XVIII) lowers exact match as expected, but the lost answers become *abstentions*, not wrong answers: at 10% and 20% edge removal the wrong-answer rate stays at 0.02 and 0.04 while abstention rises to 0.13 and 0.41. The verifier-gated planner degrades gracefully under graph staleness, refusing rather than fabricating.

TABLE XVIII
KG INCOMPLETENESS ON JOIN AND MULTI-HOP QUESTIONS (792 QUESTIONS).

Edges removed	0%	10%	20%	30%
Exact match	0.939	0.845	0.552	0.518
Wrong	0.010	0.023	0.035	0.046
Abstain	0.051	0.133	0.413	0.437

d) Efficiency.: Table XIX breaks per-question latency into stages over 1,392 questions on a 6,509-entry index. Retrieval dominates (P50/P95 10.7/11.6 ms), and execution, which includes the verifier and ensemble scoring, is sub-millisecond (P50/P95 0.6/1.2 ms). Verification is not the bottleneck at this scale, and the production path for larger corpora is approximate search.

TABLE XIX
STAGE-WISE PER-QUESTION LATENCY (CPU, 1,392 QUESTIONS, 6,509 INDEX ENTRIES).

Stage	P50 (ms)	P95 (ms)
Retrieval	10.7	11.6
Execution (incl. verifier)	0.6	1.2
Total	11.3	15.2

e) Where gating hurts.: The gate is not free. On clean join questions the no-verifier planner is perfect (EM 1.00) because the gold candidate is present, whereas the verifier rejects 16 of 264 correct joins as false negatives near the decision boundary, lowering EM to 0.94 (a 6% cost). This is the price of safety on easy positives, and HybridQA (main paper) is the macro case where gating loses outright. Raising τ_v trades this cost against bad-join rejection.

XII. VERIFIER FEATURE ABLATION

A natural question is whether the learned verifier merely recovers one obvious rule, such as numeric-suffix agreement. We retrain it on single-feature subsets. On the synthetic graded-corruption split (Table XX) no subset matches the full model: digit agreement reaches AUROC 0.70, edit-ratio 0.76, prefix overlap 0.67, and embedding only 0.52, while the full fusion reaches 0.82. The verifier learns a combination, not a single cue, and on this digit-preserving generator even digit agreement alone is far from sufficient. On the real ICD-10 hierarchy task (Table XXI) the picture differs and matches the main paper: the prefix and code-structure features alone already reach the verifier's recall@1 (0.826/0.729 all/drift), because hierarchy drift *is* a prefix relation, and the embedding adds only a small drift gain (full 0.746). Where a structural shortcut exists a single feature suffices, and where it does not the fusion is necessary.

TABLE XX
SYNTHETIC VERIFIER TRAINED ON FEATURE SUBSETS, HELD-OUT DISCRIMINATION. NO SINGLE FEATURE MATCHES THE FULL MODEL.

Feature subset	Accuracy	AUROC
Digit agreement	0.678	0.699
Edit ratio	0.683	0.757
Prefix overlap	0.637	0.669
Embedding	0.566	0.517
Full	**0.745**	**0.820**

TABLE XXI
ICD-10 VERIFIER TRAINED ON FEATURE SUBSETS, RECALL@1. THE STRUCTURAL (PREFIX/CODE) FEATURES CARRY THE HIERARCHY SIGNAL, AND THE EMBEDDING ADDS A SMALL DRIFT GAIN.

Feature subset	recall@1 (all)	recall@1 (drift)
Prefix overlap	0.826	0.729
Edit ratio	0.805	0.644
Embedding	0.754	0.458
Code structure	0.826	0.729
Full	**0.826**	**0.746**

XIII. VERIFIER FAILURE ANALYSIS

Table XXII characterizes where the verifier errs. On BIRD value linking the errors are *conservative*: at τ_v=0.5 the false-positive rate is only 0.4% (it rarely accepts a wrong value), and the false-negative rate is 7.2% (it sometimes rejects the gold value, which then shows up as an abstention rather than a wrong answer). On the ICD-10 join the dominant error is different and more instructive. Of the 41 recall@1 misses on 236 test records (17%), 18 are *sibling-code confusions* where the chosen wrong node shares the gold's three-character category, exactly the hard case (`R508` vs. `R509`) that a prefix rule also struggles with. The verifier moreover *commits* most of these (38 of 41 score above τ_v) rather than abstaining, and it rejects the gold pair in 18% of cases. This is the candid weakness behind the tie against the prefix rule: near-duplicate hierarchical codes are where both methods fail, and the verifier is overconfident there.

TABLE XXII
VERIFIER FAILURE ANALYSIS. BIRD ERRORS ARE CONSERVATIVE, WHILE ICD ERRORS ARE DOMINATED BY SIBLING-CODE CONFUSION THAT THE VERIFIER COMMITS RATHER THAN ABSTAINS ON.

ICD-10 join (236 test records, recall@1)	
recall@1 misses	41 (17%)
sibling-code confusion	18
committed a wrong join	38
safely abstained	3
gold pair rejected (FN)	42 (18%)
BIRD value linking (held-out pairs, τ_v=0.5)	
false negatives (gold rejected)	28 (7.2%)
false positives (wrong accepted)	72 (0.4%)

XIV. SCALABILITY OF THE MULTI-INDEX RETRIEVER

We stress the retriever as the synthetic corpus grows from 600 to 9,600 accounts (Table XXIII). Index size, build time, and per-question latency all grow linearly with the corpus: at $\sim$104k index entries the pure-Python linear-scan retriever reaches a P50 latency of 206 ms, up from 11 ms at 6.5k entries, a 19$\times$ rise for a 16$\times$ corpus. The verifier's per-pair cost is constant and stays negligible, so retrieval, not verification, is the scaling bottleneck. This linear growth confirms the cost model in the main paper and locates the crossover near 10^5 entries where the production path, approximate nearest-neighbor search and index sharding, replaces the exact scan.

TABLE XXIII
MULTI-INDEX SCALABILITY ON THE SYNTHETIC CORPUS (DETERMINISTIC VERIFIER, 150-QUESTION LATENCY SAMPLE). INDEX SIZE, BUILD TIME, AND QUERY LATENCY GROW LINEARLY WITH CORPUS SIZE.

Accounts	Questions	Entries	Build (s)	Latency P50/P95 (ms)
600	1,392	6,509	0.08	10.7/11.5
1,200	2,784	13,001	0.17	22.1/25.3
2,400	5,567	25,985	0.35	45.0/51.3
4,800	11,135	51,953	0.73	92.1/103.9
9,600	22,272	103,889	1.48	206.3/226.4

XV. RECORD-LINKAGE BASELINE, COMPOSITE KEYS, STRUCTURAL CONTEXT, AND RISK-CONTROLLED SELECTIVE PREDICTION

We add analyses that reviewers requested. The record-linkage baseline, composite-key, and risk-controlled selective-prediction studies run on the deterministic synthetic core. The structural-context study runs on the real ICD-10 layer, where the code hierarchy supplies genuine 1-hop context.

a) Classical record-linkage baseline.: We benchmark a Fellegi–Sunter probabilistic matcher [1], the canonical non-neural record-linkage model, on the same features and pairs as the learned verifier. We binarize each join feature into a field-agreement indicator, estimate the agreement probability under a match (m) and a non-match (u) from the labelled training pairs with Laplace smoothing, and score a pair by the summed log-likelihood ratio

$$s_{\mathrm{FS}} = \sum_j \mathbb{1}[a_j] \log \frac{m_j}{u_j} + \mathbb{1}[\neg a_j] \log \frac{1-m_j}{1-u_j},$$

where a_j denotes agreement on field j. Fellegi–Sunter is a strong, training-light baseline (AUROC 0.753), and the learned MLP verifier on the identical features improves on it by +0.067 AUROC (Table XXIV). The gain is real but modest, which is the honest reading. A learned matcher buys interaction structure over the continuous features that the agreement-binarized likelihood ratio discards, and the strongest agreement fields it recovers (containment, full-digit, and digit-suffix match) are exactly the numeric-key signals the generator preserves.

TABLE XXIV
FELLEGI–SUNTER PROBABILISTIC RECORD LINKAGE VERSUS THE LEARNED MLP VERIFIER ON THE SYNTHETIC EVAL PAIRS, IDENTICAL FEATURES. THE LEARNED MATCHER IMPROVES AUROC BY A MODEST, REAL MARGIN.

Matcher	AUROC
Fellegi–Sunter (log-likelihood ratio)	0.753
Learned MLP verifier	**0.820**

b) Composite (multi-attribute) keys: AND/OR composition and conflicts.: The verifier scores one pair, but a composite join over a two-attribute key is scored by combining the per-attribute match probabilities. A probabilistic AND (their product, accepting above τ_v^2) suits keys where both components must agree, and a probabilistic OR $(1-(1-s_1)(1-s_2)$, accepting above τ_v) suits alternate keys where either component identifies the record. We evaluate positives (both attributes drift yet agree), conflicts (one attribute agrees, one does not), and easy negatives (neither agrees). The product ranks composite matches well (AUROC 0.934), far above single-attribute scoring (AUROC 0.737), which cannot tell a positive from a conflict because it never inspects the disagreeing attribute (Table XXV). The two operators encode opposite key semantics on the conflict cases, and this is the central point: AND rejects a conflict (accept rate 0.018), as a composite key should, while OR treats one matching component as sufficient (accept rate 0.60 on conflicts, recall 0.91 on positives), as a many-to-one or alternate key should. Both recover drifted positives that exact composite matching almost entirely misses (0.3% recall). The composition operator is therefore a modeling choice that encodes the key's semantics, not a fixed recipe, and the per-attribute uncertainties compose into a join-level abstention signal either way.

TABLE XXV
COMPOSITE TWO-ATTRIBUTE KEY JOINS UNDER DRIFT, WITH AND (PRODUCT) AND OR COMPOSITION. AND REJECTS CONFLICTS (ONE COMPONENT DISAGREES), AS A COMPOSITE KEY SHOULD, WHILE OR ACCEPTS A SINGLE MATCHING COMPONENT, AS AN ALTERNATE OR MANY-TO-ONE KEY SHOULD. BOTH RECOVER DRIFTED POSITIVES THAT EXACT COMPOSITE MATCHING MISSES.

Method	AUROC	Recall$^+$	Conflict accept
Exact composite	n/a	0.003	0.000
Single-attribute verifier	0.737	n/a	n/a
Per-attribute OR	0.886	0.910	0.600
Per-attribute AND (product)	**0.934**	0.428	0.018

c) Risk-controlled selective prediction.: A reviewer asked whether conformal or risk-controlling selective classification [2], [3] should replace deep-ensemble abstention. We find the two are complementary rather than competing. On a held-out calibration split we select, for each target risk α, the largest accepted set whose calibration error is at most α, then report the achieved coverage and error on a disjoint test split (Table XXVI). The realized test error tracks the target, holding at $\alpha \in \{0.02, 0.05\}$ with a small finite-sample overshoot at $\alpha = 0.10$. Conformalizing the *ensemble* score rather than a single member lowers the realized risk at the strictest target (0.007 versus 0.013 at $\alpha = 0.01$) and buys more coverage at the loosest (0.424 versus 0.399 at $\alpha = 0.10$). Conformal selection supplies the calibrated operating point that the ensemble's heuristic entropy threshold lacks, and the ensemble supplies the better-ranked confidence score that conformal selection thresholds. The low coverage at strict targets is itself informative. Guaranteeing a sub-5% join error under heavy drift means accepting only the most confident quarter of pairs and abstaining on the rest, which is precisely the safety-coverage trade-off the verifier exists to expose.

TABLE XXVI
SPLIT-CONFORMAL SELECTIVE PREDICTION. COVERAGE AND ACHIEVED TEST RISK AT EACH TARGET RISK α, CONFORMALIZING A SINGLE MEMBER VERSUS THE FULL ENSEMBLE. REALIZED RISK TRACKS α, AND THE ENSEMBLE SCORE GIVES TIGHTER RISK AT STRICT TARGETS AND MORE COVERAGE AT LOOSE ONES.

	Single member		Ensemble	
Target α	Cov.	Risk	Cov.	Risk
0.01	0.212	0.013	0.197	0.007
0.02	0.212	0.013	0.197	0.007
0.05	0.244	0.044	0.236	0.034
0.10	0.399	0.110	0.424	0.113

d) Lightweight structural context.: A reviewer asked whether adding lightweight structural context beyond the pairwise features improves discrimination or calibration without sacrificing speed. We test this on the real ICD-10 layer, where the code hierarchy supplies genuine structure: the shared three-character category (the 1-hop parent in the ICD tree), the shared chapter letter, code-length specificity agreement, and a sibling indicator. We compare the base six-feature verifier against the base plus these four structural features on a controlled split, where the training pairs and ensemble seeds are identical and only the feature map differs. Structural context improves both discrimination and calibration at no measurable cost: AUROC rises from 0.834 to 0.855 and expected calibration error falls from 0.064 to 0.039, while per-pair feature extraction stays at about $6\,\mu$s (the structural features are constant-time string operations), Table XXVII. Top-1 recall barely moves ($+0.4$ points) and sibling-code confusions drop by only one. The honest reading is that 1-hop hierarchy context sharpens the score and its calibration but does not resolve sibling confusion, because siblings share the very category the context encodes, so disambiguating them needs finer structure than a single hop. This answers the question in the affirmative for calibration and ranking, and it scopes the open part precisely.

XVI. NORMALIZATION BASELINES, RECALIBRATION, REGIME DETECTION, AND ROBUSTNESS

We add further analyses that reviewers requested.

a) Practical normalization and fuzzy baselines.: The consolidated ICD baselines in the main text add three matchers

TABLE XXVII
LIGHTWEIGHT STRUCTURAL CONTEXT (ICD-10 HIERARCHY) ADDED TO THE PAIRWISE VERIFIER, CONTROLLED SPLIT. DISCRIMINATION (AUROC) AND CALIBRATION (ECE) IMPROVE AT NO LATENCY COST, WHILE TOP-1 RECALL AND SIBLING CONFUSION BARELY MOVE BECAUSE SIBLINGS SHARE THE ENCODED CATEGORY.

Verifier	AUROC	ECE	R@1 (all/drift)	Sib. conf.	μs/pair
Base (6 features)	0.834	0.064	0.822/0.746	19	6.2
+ structural (10)	**0.855**	**0.039**	0.826/0.746	18	6.0

TABLE XXIX
PLANNER-THRESHOLD SENSITIVITY ON REAL ICD PAIRS: PRECISION, RECALL, AND F1 OF THE ACCEPT DECISION AS τ_v VARIES. F1 IS STABLE NEAR THE OPERATING POINT.

τ_v	0.3	0.4	0.5	0.6	0.7
Precision	0.732	0.745	0.759	0.829	0.835
Recall	0.881	0.856	0.839	0.758	0.708
F1	**0.800**	0.797	0.797	0.792	0.766

common in data engineering. Domain-aware code canonicalization (stripping dots and spacing, then exact match) scores exactly as plain exact (0.644 all, 0.000 drift), which shows the ICD drift is genuine hierarchy drift, not a formatting artifact that a normalization pipeline would remove. Jaro–Winkler [4] on canonicalized codes ties the prefix rule (0.826 all, 0.729 drift), confirming that a strong off-the-shelf fuzzy matcher matches the hand-crafted rule on homogeneous hierarchy drift. Character TF-IDF cosine trails embedding (0.729 all, 0.407 drift). The learned verifier's drift-subset edge (0.746) is therefore over fairly-tuned string baselines.

b) Single-model recalibration.: Because ensemble ranking does not beat a single model, a reviewer asked whether temperature scaling or isotonic regression on one model recovers the ensemble's calibration at one-fifth the compute. It does not (Table XXVIII). Fitting temperature on a held-out calibration half leaves test ECE at 0.039, since the single model already sits near temperature 1, and isotonic regression overfits the calibration half to 0.043. Both remain above the ensemble's 0.024 on the same split, so the ensemble's calibration advantage is not recovered by a cheap post-hoc fix here.

TABLE XXVIII
SINGLE-MODEL RECALIBRATION VERSUS THE DEEP ENSEMBLE: TEST-HALF ECE ON THE SYNTHETIC SPLIT. NEITHER TEMPERATURE SCALING NOR ISOTONIC REGRESSION CLOSES THE GAP TO THE ENSEMBLE.

Method	ECE
Single, raw	0.039
Single, temperature scaled (T=1.1)	0.039
Single, isotonic	0.043
Deep ensemble	**0.024**

c) Threshold stability on real data.: The acceptance threshold is stable on real ICD pairs, not only in the synthetic study (Table XXIX). As τ_v rises from 0.3 to 0.7, precision rises and recall falls monotonically, and accept-decision F1 stays within $[0.79, 0.80]$ for $\tau_v \in [0.4, 0.6]$, so the fixed τ_v=0.5 sits in a stable band rather than on a cliff, matching the synthetic threshold sweep.

d) Regime detector.: A per-instance regime detector is feasible, which is why the cascade's failure is the verifier's cross-regime miscalibration rather than the routing decision. A logistic classifier over six cheap surface features (token count, digit and alphabetic fractions, mean token length, code punctuation, and a single-token flag) separates identifier queries from natural-language spans, using ICD codes as identifiers and their descriptions as spans, at 0.997 five-fold cross-validated accuracy against a 0.5 majority baseline. Routing on inputs is therefore easy; the open problem is a verifier whose score is calibrated across regimes.

e) Split protocol and leakage.: The ICD train and test sets split on records. We quantify residual overlap on the held-out test records: only 1.3% share their exact billable code with a training record and 19.5% share their gold Wikidata node, while 41.1% share the coarse three-character category. The verifier cannot exploit category-level signal because it reads only relational features of a (record, node) pair and never node identity, so the modest category overlap is not a leakage path.

f) Real-data verifier training.: We elaborate how the real-data verifier pairs are built, since the protocol governs both leakage and generalization. For each training record we form one positive pair, (record, gold node), and eight negatives by *hard-negative mining*: the four KG nodes whose label-plus-code embedding is closest to the record (the distractors a retriever would actually surface) and four uniformly random nodes. Each ensemble member trains on a bootstrap resample of these pairs with a distinct hidden width, and labels are taken as given rather than smoothed, since the ICD pairing is exact at construction (a record's gold node is its true category). The deliberate label ambiguity is confined to the *synthetic* graded-corruption split, where the same-entity probability decreases with edit distance through a $\sim$50/50 boundary at two edits, which is what gives the verifier a non-degenerate operating point and the ensemble meaningful variance.

g) Retrieval noise on real data.: The retrieval-noise probe of the controlled study repeats on the real ICD corpus (Table XXX). We drop the gold KG node from the candidate pool with probability p and compare the no-verifier planner, which commits the top-scored node, against the verifier, which commits only when the top score clears τ_v. As p rises the no-verifier wrong-answer rate climbs to 1.00, because every dropped-gold record yields a silent wrong join, whereas the verifier holds its wrong rate near 0.25 and converts the rest into abstentions, reaching 0.75 at p=1. The value of gating again grows as retrieval degrades, now on real identifiers.

h) Enterprise-scale retrieval.: We demonstrate the scaling cost model empirically rather than asserting it (Table XXXI). Over N unit 384-vectors we time a FAISS exact (flat) index and an approximate inverted-file (IVF) index. Exact search grows linearly, from 0.04 ms at 10^4 to 28 ms

TABLE XXX
RETRIEVAL NOISE ON THE REAL ICD CORPUS: DROPPING THE GOLD NODE WITH PROBABILITY p. THE VERIFIER CONVERTS UNRECOVERABLE JOINS INTO ABSTENTIONS WHILE THE NO-VERIFIER PLANNER RETURNS SILENT WRONG ANSWERS.

p	0.00	0.25	0.50	0.75	1.00
No-verifier wrong	0.18	0.39	0.59	0.79	1.00
Verifier wrong	0.18	0.18	0.21	0.24	0.25
Verifier abstain	0.00	0.21	0.39	0.56	**0.75**

at 10^6 candidates, while approximate search stays under half a millisecond throughout, roughly a $60\times$ gap at 10^6 (latencies are mean per-query and hardware-dependent, but the linear-versus-flat trend is the robust point). Because the verifier scores only the few candidates the index returns, its per-join cost is constant in N, so at enterprise scale retrieval, not verification, is the bottleneck and approximate search keeps it cheap.

TABLE XXXI
MEAN PER-QUERY LATENCY OF FAISS EXACT (FLAT) VERSUS APPROXIMATE (IVF) SEARCH OVER N UNIT 384-VECTORS. EXACT SEARCH SCALES LINEARLY WITH N; APPROXIMATE SEARCH STAYS ROUGHLY FLAT.

N	10^4	10^5	10^6
Exact (flat) ms	0.036	0.344	28.041
Approximate (IVF) ms	0.016	0.012	**0.480**

XVII. REPRODUCTION

Every table above regenerates from the released harness on the same public benchmarks and seeded corpus, with tested package, model, and dataset versions and a table-to-command map given in the artifact's reproduction guide. The corruption sweep (Table VI) and noise probe (Table VIII) come from the ICD-10 join driver run at varying code-corruption fraction and description-noise level. The per-seed heterogeneous table (VII) comes from the heterogeneous-drift driver aggregated over seeds 0 to 4, and the transfer and noise numbers from the same analysis entry point. The cross-encoder result (Table IX) comes from fine-tuning a MiniLM cross-encoder, HybridQA (Table X) from the evidence-retrieval driver, BIRD value linking (Table XI) from the value-linking driver, and end-to-end EX (Table XII) from the text-to-SQL ablation. The end-to-end selective-prediction, retrieval-noise, and cost-aware-utility diagnostics (Tables XIII through XV), the threshold-sensitivity, compute-matched-uncertainty, KG-incompleteness, efficiency, and gating-cost diagnostics (Tables XVI through XIX), and the record-linkage baseline, composite-key, and risk-controlled selective-prediction analyses (Tables XXIV through XXVI), all regenerate from the dependency-light synthetic harness on the seeded corpus. The structural-context study (Table XXVII) regenerates from the ICD-10 driver, which adds the four hierarchy features and compares the base and augmented verifiers on the same split. The single-model recalibration table (XXVIII) comes from the synthetic calibration harness, and the consolidated normalization baselines, real-ICD threshold sweep (Table XXIX), regime detector, and leakage quantification come from the ICD-10 driver. The pretrained embedding model and local coder make these seeded but not bit-identical across hardware, and reported values are single-seed point estimates except where a mean$\pm$sd over seeds is given.

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